



CDA 3103

Recitation 3 – Karnaugh Maps



Logic Minimization

- Our goal is to create a minimal representation of logic functions
- This is done by:
 1. Rewrite the function as the sum of its minterms
 - Sum of products canonical form
 2. Place minterms in a Karnaugh map
 3. Use Karnaugh map to determine the minimal function

Review: Minterms

- A **minterm** is a special product of literals, in which each input variable appears exactly once.
- A function with n variables has 2^n minterms (since each variable can appear complemented or not)
- A three-variable function, such as $f(x,y,z)$, has $2^3 = 8$ minterms:

Minterm	Is true when...	Shorthand
$x'y'z'$	$x=0, y=0, z=0$	m_0
$x'y'z$	$x=0, y=0, z=1$	m_1
$x'yz'$	$x=0, y=1, z=0$	m_2
$x'yz$	$x=0, y=1, z=1$	m_3
$xy'z'$	$x=1, y=0, z=0$	m_4
$xy'z$	$x=1, y=0, z=1$	m_5
xyz'	$x=1, y=1, z=0$	m_6
xyz	$x=1, y=1, z=1$	m_7

Review: Karnaugh Maps

- A two-variable function has four possible minterms. We can re-arrange these minterms into a **Karnaugh map**.

x	y	minterm
0	0	$x'y'$
0	1	$x'y$
1	0	xy'
1	1	xy



		X	
		0	1
y {	0	$x'y'$	xy'
	1	$x'y$	xy

Review: Karnaugh Maps

- Now we can easily see which minterms contain common literals.
- Minterms on the left and right sides contain x' and x respectively.
- Minterms in the top and bottom rows y' and y respectively.

		x	
		0	1
y	0	$x'y'$	xy'
	1	$x'y$	xy

	x'	x
y'	$x'y'$	xy'
y	$x'y$	xy

Review: Karnaugh Maps

- Imagine a two-variable sum of minterms:

$$x'y' + x'y$$

- Both of these minterms appear in the left column of a Karnaugh map, which means that they both contain the literal x' .

		x
	$x'y'$	xy'
y	$x'y$	xy

- What happens if you simplify this expression using Boolean algebra?

$$\begin{aligned}x'y' + x'y &= x'(y' + y) && [\text{Distributive}] \\ &= x' \bullet 1 && [y + y' = 1] \\ &= x' && [x \bullet 1 = x]\end{aligned}$$

Review: Karnaugh Maps

- For a three-variable expression with inputs x , y , z , the arrangement of minterms is more tricky:

		xy			
		00	01	11	10
z	0	$x'y'z'$	$x'yz'$	xyz'	$xy'z'$
	1	$x'y'z$	$x'yz$	xyz	$xy'z$

		xy			
		00	01	11	10
z	0	m_0	m_2	m_6	m_4
	1	m_1	m_3	m_7	m_5

- Another way to label the K-map

		x			
		$x'y'z'$	$x'yz'$	xyz'	$xy'z'$
z		$x'y'z'$	$x'yz'$	xyz'	$xy'z'$
		$x'y'z$	$x'yz$	xyz	$xy'z$
		y			

		x			
		m_0	m_2	m_6	m_4
z		m_0	m_2	m_6	m_4
		m_1	m_3	m_7	m_5
		y			

Review: Karnaugh Maps

- With this ordering, any group of 2, 4 or 8 adjacent squares on the map contains common literals that can be factored out.
- “Adjacency” includes wrapping around the left and right sides:

		XY			
		00	01	11	10
Z	0	$x'y'z'$	$x'yz'$	xyz'	$xy'z'$
	1	$x'y'z$	$x'yz$	xyz	$xy'z$

$$\begin{aligned} & x'y'z + x'yz \\ &= x'z(y' + y) \\ &= x'z \cdot 1 \\ &= x'z \end{aligned}$$

Review: Karnaugh Maps

- We'll use this property of adjacent squares to do our simplifications.

		XY			
		00	01	11	10
Z	0	$x'y'z'$	$x'yz'$	xyz'	$xy'z'$
	1	$x'y'z$	$x'yz$	xyz	$xy'z$

$$\begin{aligned}
 & x'y'z + x'yz \\
 = & x'z(y' + y) \\
 = & x'z \cdot 1 \\
 = & x'z
 \end{aligned}$$

		XY			
		00	01	11	10
Z	0	$x'y'z'$	$x'yz'$	xyz'	$xy'z'$
	1	$x'y'z$	$x'yz$	xyz	$xy'z$

$$\begin{aligned}
 & x'y'z' + x'yz' + x'y'z + x'yz \\
 = & y'(x'z' + xz' + x'z + xz) \\
 = & y'(z'(x' + x) + z(x' + x)) \\
 = & y'(z' + z) \\
 = & y'
 \end{aligned}$$



Example K-map simplification

- ▶ Let's consider simplifying $f(x,y,z) = xy + y'z + xz$.
- ▶ First, you should convert the expression into a sum of minterms form, if it's not already.
 - ▶ The easiest way to do this is to make a truth table for the function, and then read off the minterms.
 - ▶ You can either write out the literals or use the minterm shorthand.

Example K-map simplification

- Let's consider simplifying $f(x,y,z) = xy + y'z + xz$.
- Here is the truth table and sum of minterms for our example:

x	y	z	f(x,y,z)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$\begin{aligned}f(x,y,z) &= x'y'z + xy'z + xyz' + xyz \\ &= m_1 + m_5 + m_6 + m_7\end{aligned}$$

Unsimplifying expressions

- ▶ You can also convert the expression to a sum of minterms with Boolean algebra.
 - ▶ Apply the distributive law in reverse to add in missing variables.
 - ▶ Very few people actually do this, but it's occasionally useful.

$$\begin{aligned}xy + y'z + xz &= (xy \bullet 1) + (y'z \bullet 1) + (xz \bullet 1) \\&= (xy \bullet (z' + z)) + (y'z \bullet (x' + x)) + (xz \bullet (y' + y)) \\&= (xyz' + xyz) + (x'y'z + xy'z) + (xy'z + xyz) \\&= xyz' + xyz + x'y'z + xy'z\end{aligned}$$

- ▶ In both cases, we're actually “unsimplifying” our example expression.
 - ▶ The resulting expression is larger than the original one!
 - ▶ But having all the individual minterms makes it easy to combine them together with the K-map.

Making the example K-map

- Next up is drawing and filling in the K-map.
 - Put 1s in the map for each minterm, and 0s in the other squares.
 - You can use either the minterm products or the shorthand to show you where the 1s and 0s belong.
- In our example, we can write $f(x,y,z)$ in two equivalent ways.

$$f(x,y,z) = x'y'z + xy'z + xyz' + xyz$$

				X
	$x'y'z'$	$x'yz'$	xyz'	$xy'z'$
Z	$x'y'z$	$x'yz$	xyz	$xy'z$
		y		

$$f(x,y,z) = m_1 + m_5 + m_6 + m_7$$

				X
	m_0	m_2	m_6	m_4
Z	m_1	m_3	m_7	m_5
		y		

- In either case, the resulting K-map is shown below.

				X
	0	0	1	0
Z	1	0	1	1
		y		

- You can also fill in the K-map directly from a truth table.
 - The output in row i of the table goes into square m_i of the K-map.
 - Remember that the rightmost columns of the K-map are “switched.”

x	y	z	$f(x,y,z)$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0

[illegible]

1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$f(x,y,z) = m_1 + m_5 + m_6 + m_7$$

			X
	m_0	m_2	m_6
Z	m_1	m_3	m_7
			y

Grouping the minterms together

- ▶ The most difficult step is grouping together all the 1s in the K-map.
- ▶ Make **rectangles** around groups of one, two, four or eight 1s.
- ▶ All of the 1s in the map should be included in at least one rectangle.
- ▶ Do *not* include any of the 0s.

				X
	0	0	1	0
Z	1	0	1	1
		Y		

Grouping the minterms together

- Each group corresponds to one product term. For the simplest result:
 - Make as few rectangles as possible, to minimize the number of products in the final expression.
 - Make each rectangle as large as possible, to minimize the number of literals in each term.
 - It's all right for rectangles to overlap, if that makes them larger.

			X	
	0	0	1	0
Z	1	0	1	1
	y			

Reading the MSP from the K-map

- Finally, you can find the minimal sum of products (MSP).
 - Each rectangle corresponds to one product term.
 - The product is determined by finding the common literals in that rectangle.

			X	
	0	0	1	0
Z	1	0	1	1
			y	

			X	
	$x'y'z'$	$x'yz'$	xyz'	$xy'z'$
Z	$x'y'z$	$x'yz$	xyz	$xy'z$
			y	

- For our example, we find that $xy + y'z + xz = y'z + xy$.

Practice K-map 1

- Simplify the sum of minterms, $f(x,y,z) = m_1 + m_3 + m_5 + m_6$.

			X	
Z				

			X	
	m_0	m_2	m_6	m_4
Z	m_1	m_3	m_7	m_5

Solutions for practice K-map 1

- Here is the filled in K-map, with all groups shown.
 - The blue and green groups overlap, which makes each of them as large as possible.
 - Minterm m_6 is in a group all by its lonesome.

	x			
	0	0	1	0
z	1	1	0	1
	y			

- The final MSP here is $x'z + y'z + xyz'$.

Four-variable K-maps

- ▶ We can do four-variable expressions too!
 - ▶ The minterms in the third and fourth columns, *and* in the third and fourth rows, are switched around.
 - ▶ Again, this ensures that adjacent squares have common literals.

		W		
		$w'x'y'z'$	$w'xy'z'$	
y	$w'x'y'z$	$w'xy'z$	$wxy'z$	$wx'y'z$
	$w'x'yz$	$w'xyz$	$wxyz$	$wx'yz$
	$w'x'yz'$	$w'xyz'$	$wxyz'$	$wx'yz'$
		X		Z

		W		
		m_0	m_4	
y	m_1	m_5	m_{13}	m_9
	m_3	m_7	m_{15}	m_{11}
	m_2	m_6	m_{14}	m_{10}
		X		Z

- ▶ Grouping minterms is similar to the three-variable case, but:
 - ▶ You can have rectangular groups of 1, 2, 4, 8 or 16 minterms.
 - ▶ You can wrap around *all four* sides.

Simplify $f(w, x, y, z) =$
 $m_0 + m_2 + m_5 + m_8 + m_{10} + m_{13}$

- The expression is already a sum of minterms, so here's the K-map:

		w			
		1	0	0	1
y	z	0	1	1	0
		0	0	0	0
		1	0	0	1
	x				

		w			
		m_0	m_4	m_{12}	m_8
y	z	m_1	m_5	m_{13}	m_9
		m_3	m_7	m_{15}	m_{11}
		m_2	m_6	m_{14}	m_{10}
	x				

Simplify $f(w, x, y, z) = m_0 + m_2 + m_5 + m_8 + m_{10} + m_{13}$

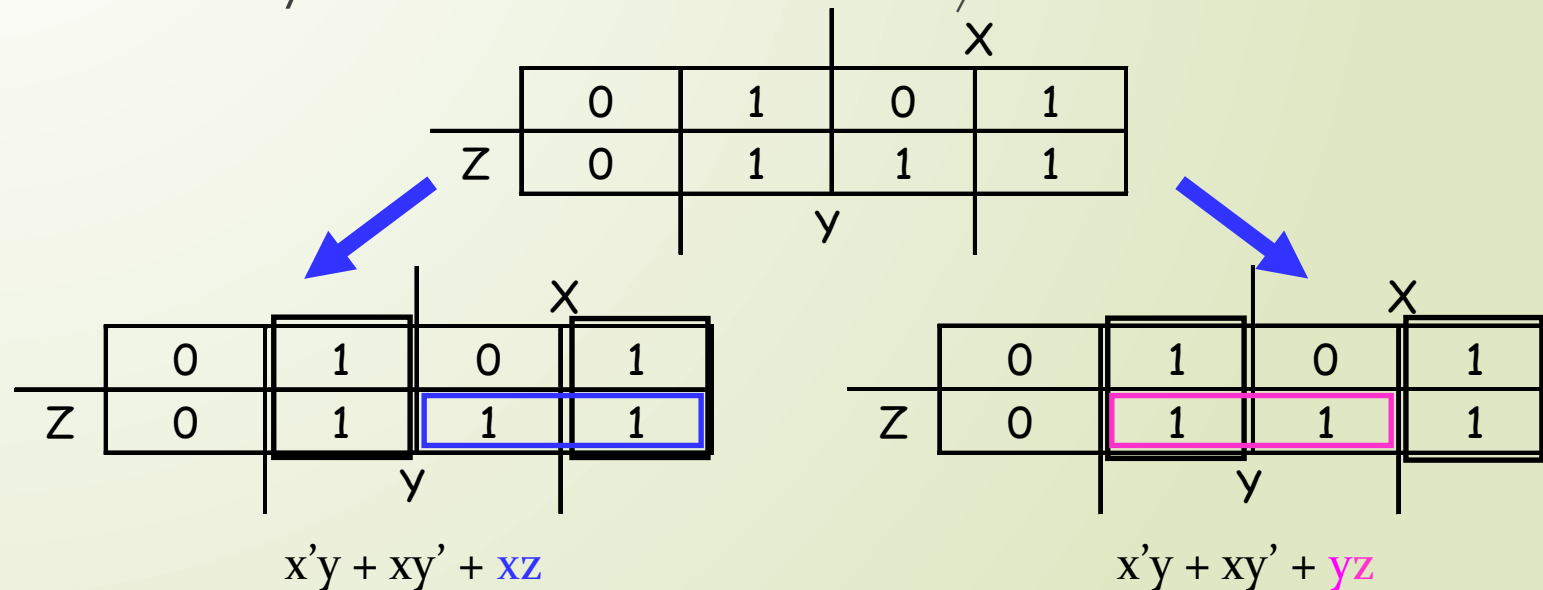
- We can make the following groups, resulting in the MSP $x'z' + xy'z$.

		w				
		1	0	0	1	
		0	1	1	0	
y		0	0	0	0	z
		1	0	0	1	
		x				

		w				
		$w'x'y'z'$	$w'xy'z'$	$wxy'z'$	$wx'y'z'$	
		$w'x'y'z$	$w'xy'z$	$wxy'z$	$wx'y'z$	
y		$w'x'yz$	$w'xyz$	$wxyz$	$wx'yz$	z
		$w'x'y'z'$	$w'xyz'$	$wxyz'$	$wx'y'z'$	
		x				

K-maps can be tricky!

- There may not necessarily be a *unique* MSP. The K-map below yields two valid and equivalent MSPs, because there are two possible ways to include minterm m_7 .



- Remember that overlapping groups is possible, as shown above.



Prime implicants

- The challenge in using K-maps is selecting the right groups. If you don't minimize the number of groups and maximize the size of each group:
 - Your resulting expression will still be equivalent to the original one.
 - But it won't be a *minimal* sum of products.

Prime implicants

- First find all of the largest possible groupings of 1s.
 - These are called the **prime implicants**.
 - The final MSP will contain a subset of these prime implicants.
- Here is an example Karnaugh map with prime implicants marked:

		w			
y		1	1	0	0
		1	1	0	0
	x	0	1	1	0
		0	0	1	1
		z			

Essential prime implicants

- If any group contains a minterm that is not also covered by another overlapping group, then that is an **essential prime implicant**.
- Essential prime implicants *must* appear in the MSP, since they contain minterms that no other terms include.
- Our example has just two essential prime implicants:
 - The red group ($w'y'$) is essential, because of m_0 , m_1 and m_4 .
 - The green group ($wx'y$) is essential, because of m_{10} .

				W	
	1	1	0	0	
	1	1	0	0	
y	0	1	1	0	Z
	0	0	1	1	
			X		



				W	
	1	1	0	0	
	1	1	0	0	
y	0	1	1	0	Z
	0	0	1	1	
			X		

Covering the other minterms

- Finally pick as few other prime implicants as necessary to ensure that all the minterms are covered.
- After choosing the red and green rectangles in our example, there are just two minterms left to be covered, m_{13} and m_{15} .
 - These are both included in the blue prime implicant, wxz .
 - The resulting MSP is $w'y' + xyz + wyz'$.
- The black and yellow groups are not needed, since all the minterms are covered by the other three groups.

		W		Z
y	X			
		0	1	
1	1	0	0	
1	1	0	0	
0	1	1	0	
0	0	1	1	

Practice K-map 2

➤ Simplify for the following K-map:

		w		
	0	0	1	0
	1	0	1	1
y	1	1	1	1
	0	0	1	0
		x		
				z

Solutions for practice K-map 2

➤ Simplify for the following K-map:

		w		
	0	0	1	0
	1	0	1	1
y	1	1	1	1
	0	0	1	0
		x		z

All prime implicants are circled.

Essential prime implicants are $x'z$, yz and wx .

The MSP is $x'z + yz + wx$.

(Including the group wz would be redundant.)

Practice K-map 3

➡ Find a MSP for

$$f(w,x,y,z) = \Sigma m(0,2,4,5,8,14,15) + d(7,10,13)$$

This notation means that input combinations $wxyz = 0111$, 1010 and 1101 (corresponding to minterms m_7 , m_{10} and m_{13}) are unused or don't cares.

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Solutions for practice K-map 3

- All prime implicants are circled. We can treat X's as 1s if we want, so the red group includes two X's, and the light blue group includes one X.

$$f(w,x,y,z) = \sum m(0,2,4,5,8,14,15) + d(7,10,13)$$

- The *only* essential prime implicant is $x'z$. The red group is not essential because the minterms (1's) in it also appear in other groups.

- The MSP is $x'z' + wxy + w'xy'$. It turns out the red group is redundant; we can cover all of the minterms in the map without it.

